From Euler to Ramanujan: Counting Integer Partitions and the Congruences Hidden in Them

Despina Ioannide
Combinatorics and Algorithm Design, Tsinghua University
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Abstract

A *partition* of n is a way of writing n as an unordered sum of positive integers; $p(n)$ counts them. We study the four problems of lecture 1 (existence, enumeration, analysis, optimization) on this one object. We count $p(n)$ with three algorithms and verify them against each other and against OEIS; we prove Euler's theorem with an explicit bijection; we prove two of Ramanujan's three congruences ($p(5n+4) \equiv 0 \bmod 5$, $p(7n+5) \equiv 0 \bmod 7$) and discover them by computer search; we show, with data, why Dyson's rank explains the first two but fails for 11 while the crank does not; and we measure how well the Hardy–Ramanujan formula predicts $p(n)$ up to $n = 10\,000$. Every number in this report was produced by code that is included with the submission, and an interactive web page lets the reader repeat each experiment. Throughout we say explicitly what is proved here, what is cited, and what is only verified numerically.

1 Introduction

Combinatorics asks four questions about a discrete structure: does an arrangement exist, how many are there, what are their properties, and which is best? For partitions the second question looks innocent. The number 4 has five partitions (4, 3+1, 2+2, 2+1+1, 1+1+1+1), and $p(100) = 190\,569\,292$. There is no closed formula for $p(n)$, yet in 1919 Ramanujan noticed, reading a table of values computed by MacMahon, that $p(4) = 5,\ p(9) = 30,\ p(14) = 135, \ldots$ are all divisible by 5. Why?

Plan. Section 2 builds the counting tools from the course (generating functions, bijections, recurrences) and compares three algorithms. Section 3 proves Euler's theorem. Section 4 is the heart of the report: the congruences, their proofs, and the combinatorial reason behind them. Section 5 studies growth. Section 6 treats the Rogers–Ramanujan identities. Section 7 describes the code and web page, and Section 8 states honestly what was proved, cited and verified.

2 Counting partitions

2.1 Generating function and Ferrers diagrams

Write $(q;q)_\infty = \prod_{m\geq 1}(1-q^m)$ and $p(0) = 1$.

Theorem 1. $\displaystyle\sum_{n\geq 0} p(n)q^n = \prod_{m\geq 1} \frac{1}{1-q^m}.$

Proof. Expand $1/(1-q^m) = 1 + q^m + q^{2m} + \cdots$. Choosing the exponent $k_m m$ from factor m means "use part m exactly k_m times". The coefficient of q^n therefore counts sequences (k_m) with

$\sum_m mk_m = n$, i.e. partitions of n. Only factors with $m \le n$ affect the coefficient of q^n, so the infinite product is a well-defined formal power series. □

Editing the product restricts the partitions: parts of size at most k give $\prod_{m\le k}(1-q^m)^{-1}$; distinct parts give $\prod_m(1+q^m)$; odd parts give $\prod_{m \text{ odd}}(1-q^m)^{-1}$.

A partition $\lambda = (\lambda_1 \ge \lambda_2 \ge \dots)$ is drawn as a *Ferrers diagram*: row i has λ_i dots. Reflecting the diagram in its diagonal gives the *conjugate* λ' (Figure 1). Conjugation is an involution on partitions of n, so *the number of partitions with largest part k equals the number with exactly k parts*: counting one family by counting a family in bijection with it, exactly the model-transformation idea of lecture 1.

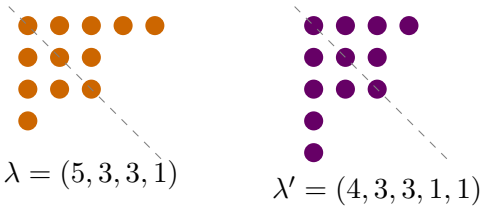


Figure 1: A partition of 12 and its conjugate. Each dot swaps its row and column.

2.2 Euler's pentagonal number theorem and a recurrence

Theorem 2 (Euler). $(q;q)_\infty = \sum_{k\in\mathbb{Z}} (-1)^k q^{k(3k-1)/2} = 1 - q - q^2 + q^5 + q^7 - q^{12} - q^{15} + \cdots$

Proof sketch (Franklin's involution). The coefficient of q^n in $(q;q)_\infty$ is the number of partitions of n into an *even* number of distinct parts minus the number into an *odd* number. Pair these up: let b be the smallest part and s the length of the initial run $\lambda_1, \lambda_1 - 1, \lambda_1 - 2, \dots$ of consecutive values. If $b \le s$, delete the bottom row and add one to each of the top b rows; if $b > s$, subtract one from each of the top s rows and add a new bottom row of length s. Each move changes the number of parts by one, keeps n, and the two moves undo each other. The pairing fails only for $k, k+1, \dots, 2k-1$ and $k+1, \dots, 2k$, which have sizes $k(3k \mp 1)/2$ and sign $(-1)^k$. □

Since $(q;q)_\infty \cdot \sum p(n)q^n = 1$, comparing coefficients gives, for $n \ge 1$,

$$p(n) = \sum_{k\ge 1} (-1)^{k+1} \Big[p\big(n - \tfrac{k(3k-1)}{2}\big) + p\big(n - \tfrac{k(3k+1)}{2}\big) \Big], \qquad p(m) = 0 \ (m < 0). \tag{1}$$

For example $p(10) = p(9) + p(8) - p(5) - p(3) = 30 + 22 - 7 - 3 = 42$.

2.3 Three algorithms

We implemented three independent ways to compute $p(n)$ (C++17, `partitions_count.cpp`):

Method	Idea	Time	Memory
Generation (brute force)	start from (n) and repeatedly take the successor in reverse lexicographic order: decrease the rightmost part larger than 1 and refill the remainder greedily with parts of that size; count the partitions visited	$\Omega(p(n))$	$O(n)$
Dynamic programming	$d[k][j] = d[k-1][j] + d[k][j-k]$: partitions of j with parts $\le k$ either avoid k or use it at least once; this is coefficient extraction in $\prod_{k \le n}(1-x^k)^{-1}$	$O(n^2)$ additions	$O(n)$ (rolling row)
Pentagonal recurrence	equation (1), about $\sqrt{n}$ terms per value	$O(n^{3/2})$ additions for a table up to n	$O(n)$ big integers

The brute-force method is feasible only for small n, because $p(n)$ grows so fast, but it is the one that makes the definition tangible. The other two need big integers: $p(1000)$ already has 32 digits (we used 128-bit integers in C++ and `BigInt` in JavaScript).

n	brute force	DP	pentagonal	n	$p(n)$ (DP = pentagonal)
20	627	627	627	100	190 569 292
30	5 604	5 604	5 604	200	3 972 999 029 388
40	37 338	37 338	37 338	500	2 300 165 032 574 323 995 027
50	204 226	204 226	204 226	1000	24 061 467 864 032 622 473 692 149 727 991
60	966 467	966 467	966 467		

Verification. The three methods agree for all $n \le 60$ (brute force) and DP equals the pentagonal recurrence for all $n \le 1000$. $p(100)$ and $p(200)$ match OEIS A000041. The JavaScript engine in the web page and an independent Python implementation reproduce the C++ table for all $n \le 1000$, and the Python/JavaScript pair agree up to $n = 10\,000$ ($p(10\,000)$ has 107 digits).

3 Euler's theorem

Theorem 3 (Euler)**.** *The number of partitions of n into distinct parts equals the number of partitions of n into odd parts.*

Proof 1 (generating functions). Using $1 + x = (1 - x^2)/(1 - x)$,

$$\prod_{m \ge 1}(1 + q^m) = \prod_{m \ge 1} \frac{1 - q^{2m}}{1 - q^m} = \frac{\prod_m (1 - q^{2m})}{\prod_{m \text{ odd}}(1 - q^m) \prod_m (1 - q^{2m})} = \prod_{m \text{ odd}} \frac{1}{1 - q^m}. \qquad \square$$

Proof 2 (Glaisher's bijection). Given a partition into odd parts, take an odd part m with multiplicity c, write c in binary, $c = \sum_i 2^{a_i}$, and replace the c copies of m by the distinct parts $2^{a_i} m$. Distinct odd m give distinct numbers because every integer factors uniquely as $2^a \cdot (\text{odd})$. The inverse splits each part $2^a m$ (m odd) into 2^a copies of m. $\square$

odd parts of 9	$\mapsto$ distinct parts	odd parts of 9	$\mapsto$ distinct parts
9	9	$3+3+3$	$6+3$
$7+1+1$	$7+2$	$3+3+1+1+1$	$6+2+1$
$5+3+1$	$5+3+1$	$3+1^6$	$4+3+2$
$5+1^4$	$5+4$	1^9	$8+1$

All eight images are different and they are exactly the eight partitions of 9 into distinct parts (9, 8+1, 7+2, 6+3, 5+4, 6+2+1, 5+3+1, 4+3+2). The code checks the equality of the two counts for every $n \le 60$ (for $n = 60$ both are 10 880).

4 Ramanujan's congruences

4.1 Discovering them by computer

Following the syllabus habit of verifying by computation, we let a program *search* for congruences: for each modulus $m \in [2, 30]$ and residue b, it tests whether $p(mn + b) \equiv 0 \pmod{m}$ for every n with $mn + b \leq 1000$ (`congruences.cpp`).

m	b	terms tested	status
5	4	200	Ramanujan
7	5	143	Ramanujan
11	6	91	Ramanujan
25	24	40	modulus 5^2 (see §4.4)

No other pair (m, b) with $m \leq 30$ survives (repeating the same test in Python up to $m = 36$ adds only $(35, 19)$); for instance $p(13n + 6)$ fails already at $n = 0$, since $p(6) = 11$. The sets of residues $p(n) \bmod m$ show the pattern at a glance (Figure 2): the column $n \equiv 4 \pmod 5$ is entirely zero. In the web page's live scan (moduli up to 36) the value $m = 35$ also appears; it is not new, because $35n + 19 \equiv 4 \pmod 5$ and $\equiv 5 \pmod 7$, so both congruences apply (Chinese remainder theorem).

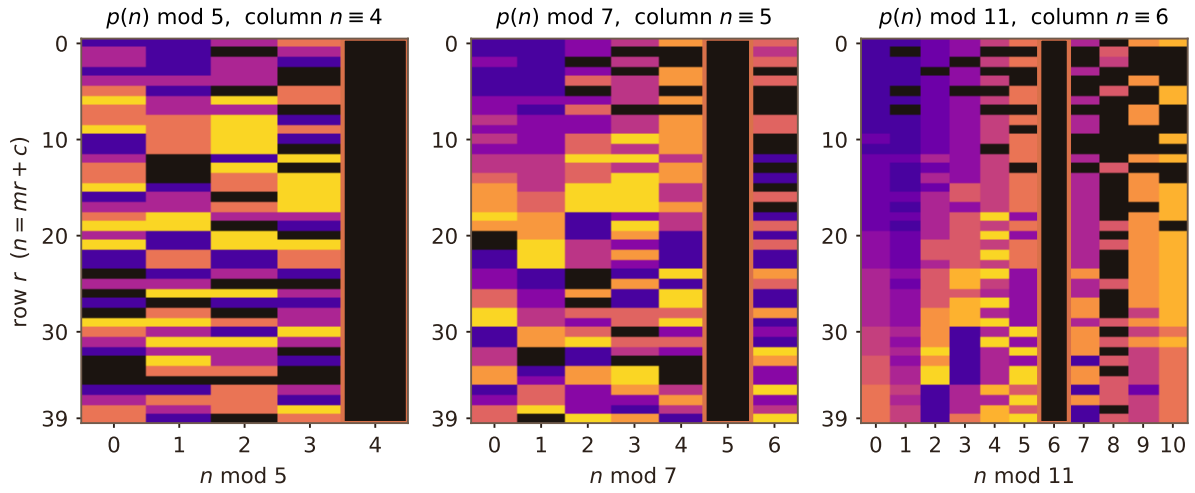


Figure 2: $p(n) \bmod m$ laid out m values per row. The dark column (outlined) contains only zeros.

4.2 Proofs for 5 and 7

Theorem 4 (Ramanujan, 1919)**.** *For all $n \geq 0$: $p(5n + 4) \equiv 0 \pmod 5$, $p(7n + 5) \equiv 0 \pmod 7$, $p(11n + 6) \equiv 0 \pmod{11}$.*

We prove the first two and cite the third.

The idea in plain words. Modulo 5, the generating function of $p(n)$ equals an explicit series (a product of four copies of $(q; q)_\infty$) times a series that only contains powers of q^5. The second factor cannot move an exponent out of its residue class mod 5, so everything depends on the coefficients of the first series at exponents divisible by 5. Those coefficients can be written as an explicit double sum, and a one-line fact about squares modulo 5 shows that each term contributing to such an exponent is already divisible by 5.

Three facts used as black boxes.

(F1) Euler's pentagonal number theorem (Theorem 2).

(F2) Jacobi's identity: $(q;q)_\infty^3 = \sum_{k\geq 0}(-1)^k(2k+1)\,q^{k(k+1)/2}$ (Hardy–Wright, Thm. 357).

(F3) For a prime ℓ: $(1-x)^\ell \equiv 1-x^\ell \pmod{\ell}$.

Proof for 5.

Step 1. *Reduce mod 5.* By (F3), $(q;q)_\infty^5 \equiv (q^5;q^5)_\infty \pmod 5$. Therefore

$$\sum_{n\geq 0} p(n)q^n = \frac{1}{(q;q)_\infty} = \frac{(q;q)_\infty^4}{(q;q)_\infty^5} \equiv (q;q)_\infty^4 \cdot \frac{1}{(q^5;q^5)_\infty} \pmod 5.$$

The last factor is a series in powers of q^5 only.

Step 2. *Expand the first factor.* Write $q\,(q;q)_\infty^4 = q\cdot(q;q)_\infty\cdot(q;q)_\infty^3$ and use (F1) and (F2):

$$q\,(q;q)_\infty^4 = \sum_{j\in\mathbb{Z}}\sum_{k\geq 0}(-1)^{j+k}(2k+1)\,q^{e(j,k)}, \qquad e(j,k) = 1 + \frac{j(3j-1)}{2} + \frac{k(k+1)}{2}.$$

Step 3. *An identity for the exponent.* Expanding both sides shows

$$24\,e = (6j-1)^2 + 3(2k+1)^2 + 20.$$

Step 4. *Squares mod 5.* Suppose $5 \mid e$. Put $x = 6j-1$ and $y = 2k+1$. Step 3 gives $x^2 + 3y^2 \equiv 0 \pmod 5$, i.e. $x^2 \equiv 2y^2$. The squares modulo 5 are $0, 1, 4$, so 2 is not a square; hence $y \equiv 0 \pmod 5$. Therefore *every term of $q(q;q)_\infty^4$ whose exponent is divisible by 5 carries a factor $2k+1$ divisible by 5.*

Step 5. *Conclude.* By Step 1, $q\sum p(n)q^n \equiv \big[q(q;q)_\infty^4\big]\cdot\big[\text{series in } q^5\big]$. The coefficient of q^{5m} on the right is a combination of coefficients of q^{5i} in $q(q;q)_\infty^4$, all divisible by 5 by Step 4. On the left it is $p(5m-1)$. Taking $m = n+1$ gives $p(5n+4) \equiv 0 \pmod 5$. □

Sanity check. $q(q;q)_\infty^4 = q - 4q^2 + 2q^3 + 8q^4 - 5q^5 - 4q^6 - 10q^7 + 8q^8 + \cdots$, and the coefficients at the exponents $5, 10, 15, 20$ are $-5, 0, 0, 0$: all divisible by 5.

Proof for 7: the same steps with different numbers.

	modulus 5	modulus 7
Step 1 (reduce)	$(q;q)_\infty^4/(q^5;q^5)_\infty$	$(q;q)_\infty^6/(q^7;q^7)_\infty$
Step 2 (expand)	$q\,(q;q)_\infty\cdot(q;q)_\infty^3$, sum over (j,k)	$q^2\big((q;q)_\infty^3\big)^2$, sum over (a,b) with weight $(2a+1)(2b+1)$
Exponent e	$1 + \frac{j(3j-1)}{2} + \frac{k(k+1)}{2}$	$2 + \frac{a(a+1)}{2} + \frac{b(b+1)}{2}$
Step 3 (identity)	$24e = (6j-1)^2 + 3(2k+1)^2 + 20$	$8e = (2a+1)^2 + (2b+1)^2 + 14$
Step 4 (squares)	2 is not a square mod 5	-1 is not a square mod 7
Consequence	$5 \mid (2k+1)$	$7 \mid (2a+1)$ and $7 \mid (2b+1)$
Step 5 (conclude)	$p(5m-1)$, $m = n+1$	$p(7m-2)$, $m = n+1$

For 7 the consequence is stronger: both factors are divisible by 7, so the coefficients at exponents divisible by 7 are divisible by 49 (for instance the coefficients of q^7, q^{14}, q^{21} in $q^2(q;q)_\infty^6$ are $0, 49, 0$).

Modulus 11. We do not reproduce a proof: the classical routes need Winquist's ten-factor identity or modular forms. This case is cited (Ramanujan 1919/1921) and verified numerically only. We did not find a short route by the same method: there $(q;q)_\infty^{10} = (q;q)_\infty\cdot((q;q)_\infty^3)^3$ involves more summation variables, and the quadratic-form argument above no longer closes (this remark is informal).

4.3 Why this column? Rank and crank

Dyson (1944) asked for a *combinatorial* reason that the $p(5n+4)$ partitions split into five equal classes. He proposed the *rank*.

Definition 5. The *rank* of a partition is its largest part minus its number of parts. The *crank* (Andrews–Garvan, 1988) is the largest part if there are no ones, and otherwise $\mu - \omega$, where ω is the number of ones and μ the number of parts larger than ω.

Cited theorems. Atkin and Swinnerton-Dyer (1954) proved Dyson's conjecture that for $n \equiv 4 \pmod 5$ and $n \equiv 5 \pmod 7$ the rank is equidistributed modulo 5, respectively 7; the crank (Andrews–Garvan, 1988) is equidistributed in all three cases $5, 7, 11$. These are not proved here.

Evidence. For $n = 4$ the five partitions have ranks $3, 1, 0, -1, -3$ and cranks $4, 0, 2, -2, -4$, which are all different modulo 5. We then enumerated *all* partitions for $n = 4, 9, \ldots, 59 \pmod 5$ and $n = 5, 12, \ldots, 54 \pmod 7$ (`rank_crank.cpp`): in every case the classes have exactly $p(n)/5$, respectively $p(n)/7$, elements, e.g. $n = 59$: $p(59) = 831\,820$ and $166\,364$ partitions in each class. Away from these n the classes are uneven; for $n = 8$ the five rank classes have sizes $4, 4, 5, 5, 4$ (Figure 3).

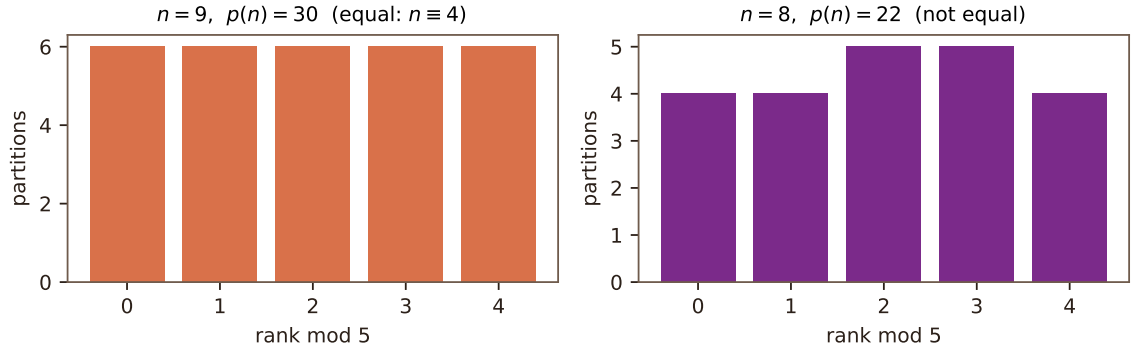


Figure 3: Rank modulo 5 for $n = 9$ (equal classes, $9 \equiv 4$) and $n = 8$ (unequal).

Mod 11 is where the rank breaks. For $n = 6, 17, 28, 39, 50$ the rank classes modulo 11 are uneven (for $n = 17$: $29, 29, 28, 26, 26, 25, 25, 26, 26, 28, 29$ out of $p(17) = 297$), but the crank classes are exactly equal (27 each for $n = 17$; 18 566 each for $n = 50$, where $p(50) = 204\,226 = 11 \cdot 18\,566$). This is Dyson's 1944 prediction that a new statistic was needed for 11, confirmed 44 years later by the crank, and reproduced by our enumeration.

4.4 Beyond 5, 7, 11: what is actually known

The slogan "only 5, 7 and 11" is true only in a precise sense, and we state it carefully.

- **Ramanujan-type congruences.** As we understand Ahlgren and Boylan (2003), a congruence $p(\ell n + \beta) \equiv 0 \pmod \ell$ valid for all n, with ℓ prime and the modulus equal to the step, can only occur for $\ell \in \{5, 7, 11\}$. [cited; the statement agrees with several secondary sources, but we did not read the full text, so the exact hypotheses (for instance a condition $\ell \geq 5$) are not checked]
- **Other steps do exist.** Ono (2000) proved, as his abstract states, that for every prime modulus larger than 3 there are infinitely many progressions $An + B$ with $p(An + B) \equiv 0$ modulo that prime; the steps A are huge. Two examples modulo 13 from the literature are $p(11^3 \cdot 13\,n + 237) = p(17303n + 237)$ (Atkin) and $p(59^4 \cdot 13\,n + 111247)$ (Ono, equation (2) of his 2000 paper). We tested them: $p(237)$, $p(17540)$, $p(34843)$ and $p(111247)$ are all divisible

by 13. (Only $n = 0$ is feasible for the second family: $n = 1$ would need $p(1.6 \times 10^8)$.) [cited; checked at small n only]

- **Prime powers.** Ramanujan conjectured, and Watson (powers of 5 and 7) and Atkin (powers of 11) proved, congruences modulo $5^a, 7^b, 11^c$. Our search *rediscovered* the first of these, a known result: $p(25n + 24) \equiv 0 \pmod{25}$ (Watson). It is genuinely stronger than the mod 5 congruence, not a consequence of it. [cited; checked for $n \le 1000$]

5 How big is $p(n)$?

A three-line argument gives an easy upper bound. For $0 < q < 1$, $p(n)q^n \le \sum_k p(k)q^k = \prod_m (1 - q^m)^{-1}$. With $q = e^{-t}$ and $e^x - 1 \ge x$,

$$\log \prod_m \frac{1}{1 - e^{-mt}} = \sum_{k \ge 1} \frac{1}{k} \cdot \frac{1}{e^{kt} - 1} \le \sum_{k \ge 1} \frac{1}{k^2 t} = \frac{\pi^2}{6t},$$

so $p(n) \le \exp\left(nt + \frac{\pi^2}{6t}\right)$, minimized at $t = \pi/\sqrt{6n}$, giving $p(n) \le e^{\pi\sqrt{2n/3}}$.

Theorem 6 (Hardy–Ramanujan, 1918)**.** $p(n) \sim \dfrac{1}{4n\sqrt{3}}\, e^{\pi\sqrt{2n/3}}$ *as* $n \to \infty$.

Rademacher (1937) turned this into an exact convergent series whose first term ($k = 1$) refines the formula; both are cited, not proved. Table 1 compares them with the exact values.

n	HR formula / exact	HR relative error	Rademacher k=1 rel. error
10	1.145341	1.5×10^{-1}	-8.9×10^{-3}
50	1.065440	6.5×10^{-2}	-7.3×10^{-5}
100	1.045714	4.6×10^{-2}	-1.8×10^{-6}
200	1.032029	3.2×10^{-2}	-9.1×10^{-9}
500	1.020095	2.0×10^{-2}	-2.4×10^{-13}
1000	1.014152	1.4×10^{-2}	-3.7×10^{-19}
10 000	1.004446	4.4×10^{-3}	—

Table 1: Accuracy of the Hardy–Ramanujan leading term (C++ for $n \le 1000$, Python/JavaScript for $n = 10\,000$).

In every case we computed ($n \le 10\,000$) the formula overestimates, with a relative error that shrinks roughly like $n^{-1/2}$. The leading Hardy–Ramanujan form is the main asymptotic term of Rademacher's $k = 1$ summand but drops a correction of relative size about $1/(\pi\sqrt{2n/3})$ that comes from differentiating; keeping the full $k = 1$ term (as our `asymptotics.cpp` does) is accurate to many digits already at moderate n (Table 1). That the overestimate persists for all n is not proved here.

6 The Rogers–Ramanujan identities

Theorem 7 (Rogers–Ramanujan identities, partition form of MacMahon and Schur)**.** *The number of partitions of n into parts that differ by at least 2 equals the number of partitions of n into parts congruent to $\pm 1 \pmod 5$.*

The sum side is easy to understand: a partition with exactly r parts differing by at least 2, after subtracting the staircase $2r - 1, 2r - 3, \ldots, 1$ from its parts in decreasing order, becomes an arbitrary partition of $n - r^2$ into at most r parts. So the generating function is $\sum_r q^{r^2}/(q;q)_r$

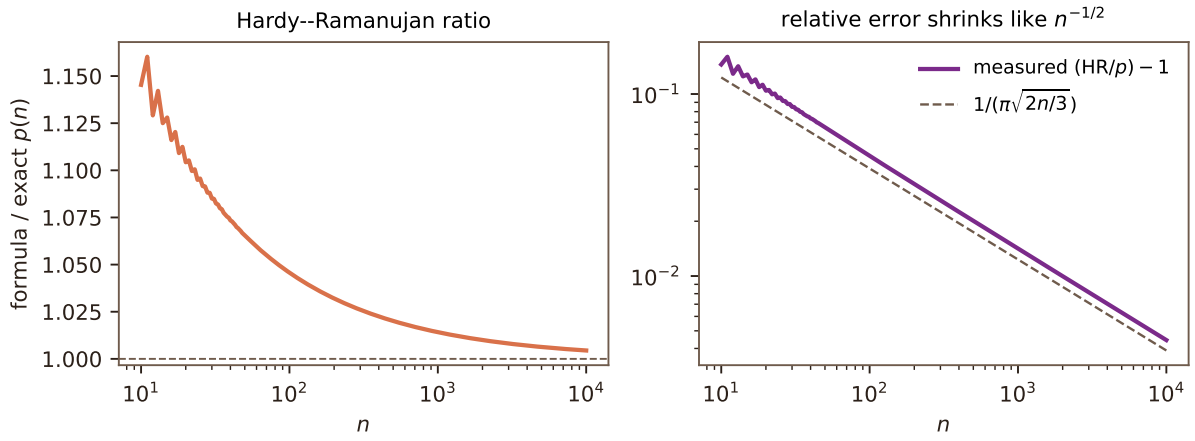


Figure 4: Left: the ratio tends to 1 from above. Right: the relative error is of the same order as $1/(\pi\sqrt{2n/3})$, in our data about 1.15 times larger (an observation, not a theorem).

(proved). Equality with the product $\prod_{m\geq 0}\left((1-q^{5m+1})(1-q^{5m+4})\right)^{-1}$ is the deep part and is cited.

Example $n = 9$. Parts differing by at least 2: 9, 8+1, 7+2, 6+3, 5+3+1 (5 partitions). Parts in $\{1, 4, 6, 9, \dots\}$: 9, $6{+}1^3$, $4{+}4{+}1$, $4{+}1^5$, 1^9 (5 partitions). The program checks equality for all $n \leq 60$ (e.g. $n = 20$: 31, $n = 40$: 374, $n = 60$: 2785). The modulus 5 here and in Ramanujan's congruence is not a coincidence: both live in the modular world of level 5.

7 The interactive explainer and reproducibility

The submission includes a static web page (HTML, CSS and JavaScript, no build step), built to be added to my personal portfolio website. Its chapters repeat this report: a Ferrers-diagram builder with an animated conjugation; a "race" between the three algorithms with their work counted live; the wall of $p(n)$ mod m with a slider for $m = 2, \dots, 36$ and a live pattern hunt; and the growth plot up to $n = 10\,000$. All values come from exact `BigInt` arithmetic in the browser. On load the page compares its own $p(n)$, $n \leq 1000$, and its congruence scan with the table exported by the C++ programs, and shows the result.

Reproducing. Programs (g++ 13, `-O2 -std=c++17`): `partitions_count`, `congruences`, `rank_crank`, `rogers_ramanujan`, `asymptotics`, `export_verified`; each prints a Markdown table and aborts if an internal check fails. `make_figures.py` regenerates the figures and cross-checks the C++ table in Python.

8 Reflection and honesty about what is proved

Result	Status	Notes
Generating function, restricted versions, Euler's theorem (two proofs)	proved here	elementary
Pentagonal theorem, recurrence	sketch + derivation	Franklin's involution sketched
Congruences mod 5 and mod 7	proved here	uses pentagonal theorem and Jacobi's identity (cited)
Congruence mod 11	cited, verified numerically	no short proof
Rank equidistribution, crank	cited, verified numerically	$n \leq 59 \pmod 5$, $\leq 54 \pmod 7$, $\leq 50 \pmod{11}$
$p(n) \leq e^{\pi\sqrt{2n/3}}$	proved here	three lines
Hardy–Ramanujan, Rademacher	cited, compared numerically	$n \leq 10\,000$
Rogers–Ramanujan identity	sum side proved; equality cited	checked $n \leq 60$
Ahlgren–Boylan, Ono, Watson	cited	two literature congruences verified at small n

Limits. Numerical verification on a finite range is evidence, not proof; the report says so wherever it matters. We did not read the full text of the Ahlgren–Boylan and Ono papers. The relative-error trend in Figure 4 is an observation.

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